

Vector Quantization for Uncertainty Quantification: The Quantized Infinitesimal Jackknife

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Abstract. Nonparametric confidence intervals (CIs) for an estimate $\hat{T}$ are usually obtained by the bootstrap: recompute the estimator on B resampled datasets and read the interval from the replicates. The cost is B evaluations of the estimator, which is prohibitive when one evaluation is an optimization. We show that a vector quantizer fitted to the data gives a much cheaper route to the same interval when the estimator’s influence function has no closed form. The Quantized Infinitesimal Jackknife (QIJ) works in two stages. A quantizer of the data supplies a rough estimate of the influence function from a few hundred evaluations on the prototypes. A second, one-dimensional quantizer groups the data by that estimate into bins inside which the influence varies little, and the estimator is differenced along each bin’s mass on the full data. Propagating the known multinomial fluctuation of the bin masses through these differences gives the variance and skewness of the sampling distribution, and a BCa-form interval. Nothing is resampled. On two synthetic distributions (Pareto; multivariate t in ten dimensions) and two astronomical datasets (the galaxy Fundamental Plane; a stellar initial mass function fitted by constrained maximum likelihood), QIJ’s interval matches the bootstrap’s coverage on twelve of thirteen estimands, with 55 to 334 full-data-evaluation equivalents against the bootstrap’s 2000, and reports the share of the variance it measured directly on every estimand.

Keywords: vector quantization · uncertainty quantification · confidence intervals · influence functions · bootstrap

1 Introduction

Vector quantization (VQ) compresses a dataset $X = \{x_i \in \mathbb{R}^d\}_{i=1}^N$ into $M \ll N$ prototypes $W = \{w_j\}_{j=1}^M$ to facilitate signal encoding, clustering, density representation [9], and, through self-organizing maps and their descendants, topology-

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preserving visualizations. In this work, we assign VQ the task of constructing confidence intervals for an arbitrary parameter estimate computed from X .

The bootstrap [4] is the gold standard for nonparametric interval estimation: recompute the estimator T on B resampled datasets, with B in the hundreds to thousands, and infer CIs from the replicated sampling distribution of T . When a single evaluation of T is expensive, e.g., the result of an optimizer-based fit, a profile likelihood, or an ensemble fit, those B evaluations dominate the analysis. Since a VQ represents a large dataset compactly, one might expect its prototypes to stand in for the N sample points during resampling; this works well for point estimates, but not for interval estimation. That is, in terms of uncertainty quantification, the quantizer prototypes are not just a smaller dataset from which to resample. Quantization suppresses precisely the sampling noise that interval estimation requires. We can, however, recover a notion of sampling variability by studying how resampling variance interacts with a VQ partition: there is variation in *how many* points land in each region, and also in *which* points represent each region. The VQ objective minimizes the second part by construction (if the VQ is trained properly); in this work we exploit multinomial models of the first to obtain CIs.

To achieve this, we replace explicit resampling with perturbation. Under bootstrap resampling, the counts $(n_1, \dots, n_M)$ of data in each region fluctuate according to a multinomial covariance. Importantly, no simulation is required to learn it. The only unknown is how sensitive the estimator is to each changing count, and that is a derivative: perturb one region’s share of the data, re-evaluate T , and record the response. Our Quantized Infinitesimal Jackknife (QIJ) does exactly this, via $\mathcal{O}(M)$ deterministic evaluations of T in total, and propagates the counts’ known covariance through these responses (the delta method), yielding the variance and skewness of T ’s sampling distribution. The result is a VQ-based confidence interval with no resampling at any stage (Sect. 2).

The perturbation route needs one thing the quantizer does not provide: bins inside which the estimator’s influence is nearly constant. The variance of an estimator splits into the part explained by how many points land in each bin and the part left by which points those are. Perturbing bin masses measures the first part exactly. Nothing measured along bin masses can recover the second, because the average response of a bin says nothing about the spread inside it. With bins that are the quantizer’s own receptive fields, the second part is small only where the influence happens to be flat inside each cell, and on the multivariate t and Fundamental Plane estimators of Sect. 4 it is more than half of the variance. The bins must therefore follow the influence, and the influence is what we do not know.

The observation that makes this possible is that the influence function is a derivative of the estimator, and a derivative can be taken on a coarse version of the data. Evaluating the estimator on the quantizer’s prototypes as a weighted dataset, and again with each prototype’s weight nudged, gives an estimate of the influence at every prototype

from a few hundred cheap evaluations. A smooth model through those values gives an estimate at every data point. Grouping the data by that estimate gives bins that follow the influence before any full-data evaluation is spent. The finite differences on the full data then measure the between-bin part, and the same differences say where the grouping was too coarse and should be refined. What remains inside the bins is predicted from the model, and the two parts add up to the variance estimate. The decomposition is an algebraic identity on any partition (Sect. 2.2), and it is what lets us report how much of the variance was measured and how much was predicted.

Related work. The infinitesimal jackknife originates with Jaeckel [14]; Efron [5] connects it to the bootstrap. Wager et al. [23], similarly motivated by an estimator too expensive to resample, apply it to random forests, though their derivative is taken with respect to individual observation weights, whereas ours is taken with respect to Voronoi masses. The Bag of Little Bootstraps [16] and m -out-of- n subsampling [2] reduce the size of each resample but still resample, and their approximation error is asymptotic and unobserved; QIJ never resamples, and its approximation error is a reported quantity. The influence-function machinery we rely on is classical robust statistics [13, 22]. **Influence functions computed by differentiating a fitted model, rather than resampling it, are used in machine learning to trace a prediction back to training points [17] and to approximate cross-validation [8]; both need the model’s Hessian, which we do not have. The ABC intervals of DiCiccio and Efron [3] build BCa-form intervals from analytic derivatives of the estimator, which is where our interval’s form comes from. Coresets [7] compress a dataset so that a fit on the summary approximates the fit on the data; our prototype stage uses the summary for a derivative, not a fit, and the two need not agree in value. Learning vector quantization with an adapted metric [12, 21] bends the quantizer toward the quantity of interest, which is what our second quantizer does with the influence itself; and quantizing by an estimated gradient rather than by position is the idea behind gradient-based dimension reduction [24].**

2 The Quantized Infinitesimal Jackknife

2.1 Setup: Influence Functions and Bin Influences

Let $X = \{x_i\}_{i=1}^N$ be iid from F , let $\hat{F}$ be its empirical distribution, and let T be a functional on distributions, so that the estimand is $\theta = T(F)$ and the estimator is its plug-in $\hat{T} = T(\hat{F})$. The influence function of T at $\hat{F}$ is

$$\psi(x) = \lim_{\varepsilon \rightarrow 0} \frac{T((1 - \varepsilon)\hat{F} + \varepsilon \delta_x) - T(\hat{F})}{\varepsilon}, \quad (1)$$

the derivative of T in the direction of a point mass at x , i.e. how much the estimate moves per unit of probability placed at x ; it satisfies $\sum_i \psi(x_i) = 0$. The

classical linearization $\hat{T} - \theta \approx \frac{1}{N} \sum_i \psi(x_i)$ [18, 13] underlies both the jackknife and the bootstrap, and gives $\text{Var}(\hat{T}) \approx \mathbb{E}[\psi^2]/N$.

Two vector quantizers appear below. The $\mathcal{X}$ -VQ quantizes the data: its prototypes are w_j , each with a receptive field RF_j of n_j points and mass $p_j = n_j/N$. The $\mathcal{I}$ -VQ quantizes an estimate of the influence itself, and its cells, which we call bins, are indexed by k . The construction that follows applies to any set K of data points, receptive field or bin. The influence of a set K with n_K points is the derivative of T with the point mass replaced by an equal spread of mass over the points of K :

$$U_K = \lim_{\varepsilon \rightarrow 0} \frac{T\left((1 - \varepsilon)\hat{F} + \frac{\varepsilon}{n_K} \sum_{i \in K} \delta_{x_i}\right) - T(\hat{F})}{\varepsilon} = \frac{1}{n_K} \sum_{i \in K} \psi(x_i). \quad (2)$$

The perturbation raises the set's mass by ε at the expense of the others, reweighting every point inside it identically (never individually). Since the derivative is linear in the direction perturbed, the response is the average of the individual responses $\psi(x_i)$, which is the second equality. **For a bin we write U_k and call it the bin influence.**

Eq. (2) is the infinitesimal jackknife (IJ) of [14] with its perturbation space restricted to the masses of M sets instead of N observation weights. At $M = N$ each set holds a single point and $U_K = \psi(x_K)$; at $M = 1$ the single set gives $U = 0$. **When ψ has a closed form, the U_K are averages of known numbers. This paper is about the other case, in which T is available only as a program: one *estimator evaluation* is a call $T(X, \omega)$ on the data with non-negative weights ω summing to N , and U_K is obtained by finite differences of that call (Sect. 2.3).**

The counts obtained under resampling are jointly multinomial: redistributing N draws into a fixed partition gives $\text{Cov}(n_k, n_l) = N(p_k \delta_{kl} - p_k p_l)$, with no simulation required to learn this covariance structure. Propagating this known fluctuation through the derivatives via the delta method gives the variance estimate

$$V_{\text{btw}} = \frac{1}{N} \sum_{k=1}^M p_k U_k^2, \quad (3)$$

using the fact that $\sum_k p_k U_k = 0$. The same quantities supply the third moment of the influence distribution, from which the skewness correction of the BCa interval [6] follows with no additional evaluations; the median-bias constant z_0 , conventionally taken from bootstrap replicates, is set to zero.

2.2 The Variance Decomposition

With U_k as above and $\text{Var}_k(\psi) = \frac{1}{n_k} \sum_{i \in k} (\psi(x_i) - U_k)^2$ the spread of influence among the points inside bin k ,

$$\underbrace{\frac{1}{N} \sum_{i=1}^N \psi(x_i)^2}_{N V_{\text{tot}}} = \underbrace{\sum_{k=1}^M p_k U_k^2}_{N V_{\text{btw}}} + \underbrace{\sum_{k=1}^M p_k \text{Var}_k(\psi)}_{N V_{\text{win}}}. \quad (4)$$

This is the between/within variance decomposition applied to the influence values, and it holds *algebraically* on any dataset and any partition. The left side contains no partition information, while both right-side terms depend on it strongly. **The identity also states the obstacle that shapes the rest of the method.** The bin influences U_k , however exactly they are obtained, say nothing about V_{win} : averages over a set of points do not determine the spread inside the set. With bins that are the receptive fields of the $\mathcal{X}$ -VQ, V_{win} is small only where ψ happens to vary little inside each Voronoi cell; on the multivariate t and Fundamental Plane estimands of Sect. 4, receptive fields at $M = 40$ –50 leave 54–80% of the variance inside them, and no number of evaluations along their masses can recover it. Without a closed form for ψ , the only route is to build bins inside which ψ is nearly constant, which requires knowing its shape before the finite differences are taken. The $\mathcal{X}$ -VQ supplies that shape cheaply (Sect. 2.4) and the $\mathcal{I}$ -VQ then makes V_{btw} nearly the whole of V_{tot} (Sect. 2.5).

2.3 Finite Differences Along a Bin's Mass

Let K be a bin with mass p . Raising its mass by a fraction δ of itself and lowering every other point's weight in proportion gives weights $\omega_i = (1-t) + t \mathbf{1}\{i \in K\}/p$ with $t = \delta p/(1-p)$; they sum to N , every weight stays non-negative for $\delta \leq 1$, and the direction is the multinomial one of (3). Write $T(t)$ for the estimator at these weights, with $T(0) = \hat{T}$ shared by every bin, and define U_K as the derivative with respect to t . The step is a fraction of the bin's own mass rather than a fixed amount of mass, so that a small bin is not perturbed proportionally more than a large one; with a fixed absolute step, a bin holding one percent of the data has its mass more than doubled, and the difference is a chord across a large change rather than a derivative. The derivative carries a truncation error from the step and a rounding error from the estimator's accuracy η , the relative size of the part of one evaluation that does not reproduce. Balancing the two gives $\delta = (3\eta)^{1/3}$ for the central stencil $U_K = [T(t) - T(-t)]/(2t)$, which also yields the second difference $\Delta^2 T_K = [T(t) - 2T(0) + T(-t)]/t^2$, and $\delta_f = 2\sqrt{\eta}$ for a single forward step $U_K = [T(t) - T(0)]/t$. The user declares η : machine precision for a closed form, the solver's tolerance for an iterative estimator.

2.4 The $\mathcal{X}$ -VQ and the Initial Influence Estimate

The shape of ψ can be learned cheaply because ψ is a derivative of T , and derivatives can be taken on a coarse version of the data. Fit the $\mathcal{X}$ -VQ with $M_{\mathcal{X}}$ prototypes (k -means centroids; $M_{\mathcal{X}}$ is set below) to the data in whitened coordinates. Evaluate T on the prototypes as a weighted dataset of $M_{\mathcal{X}}$ rows, each prototype carrying its mass, and then once more per prototype with that prototype’s mass raised by the fraction δ_f and the others lowered in proportion. The differences, mass-centred, are the *prototype influences* I_j : the derivative of T at the quantized data along prototype j ’s mass. They are influence values of the quantized distribution, not of the data, and the two can differ in scale. Centroids thin the tails of a distribution, so on the multivariate t the estimator returns a degrees-of-freedom value of about 150 on the prototypes against 5.6 on the full data of the same draw. What the next stage needs from the I_j is their level structure, which points are high and which are low, and Fig. 1 reports how well that structure agreed with the true influence. The cost is $1 + M_{\mathcal{X}}$ evaluations on $M_{\mathcal{X}}$ rows.

A Gaussian process with an affine mean and a Matérn-3/2 kernel is passed through the pairs (w_j, I_j) [20]. Its width and the ratio of its noise to its signal variance are set by maximum marginal likelihood, with the noise ratio profiled at each width, and the smallest width allowed equal to the median distance between prototypes that the $\mathcal{X}$ -VQ’s connectivity graph joins. The prototype influences carry evaluation noise: each is a difference of two evaluations of accuracy η divided by a step t_j , so its noise has standard deviation about $\sqrt{2}\eta|T|/t_j$. Left free, the likelihood on a noisy estimator drives the noise ratio to zero and interpolates that noise as signal, and the model then reports the resulting wiggle as variation inside the bins. The noise ratio is therefore not allowed below the level the declared η implies. For a closed-form estimator that level is far below the search’s floor and nothing changes; for the optimizer-based estimator of Sect. 3 it is what makes the model’s uncertainty honest. The posterior mean at a data point is the *initial influence estimate* $\hat{\psi}_0(x_i)$, and the posterior standard deviation is its uncertainty σ_i , small where prototypes are dense and large in the tails. The affine mean matters: a zero-mean kernel model returns to zero beyond the outermost prototype, and on heavy-tailed data the most influential points lie exactly there. No estimator evaluations are spent here.

How many prototypes. The first stage costs about $M_{\mathcal{X}}^2$ rows of evaluation and the finite differences of the second about $(1 + 2q M_{\text{ref}})N$ rows for q outputs, with M_{ref} the bin count of Sect. 2.5. We give the $\mathcal{X}$ -VQ the same row budget as the finite differences it serves,

$$M_{\mathcal{X}} = \left\lceil \sqrt{(1 + 2q M_{\text{ref}}) N/2} \right\rceil, \quad (5)$$

floor 20, cap $N/2$: 188 prototypes for a one-output estimator at $N = 2000$, 371 for the four-output Fundamental Plane and 414 for the five-output initial mass function. This is a budget rule, not an accuracy rule. No rule known before running guarantees that the prototypes resolve an unseen estimator's influence; Sect. 4.2 tests this on a tail probability in ten dimensions. The cost of the whole method is known before running, about twice the finite differences, against BN rows for the bootstrap; the diagnostics of Sect. 2.6 say afterwards whether the $\mathcal{X}$ -VQ was fine enough.

2.5 The $\mathcal{I}$ -VQ and Gain-Driven Refinement

Evaluate T on the full data once. Then quantize the values $\hat{\psi}_0(x_i)$ with a one-dimensional k -means into $M_{\mathcal{I}}$ bins, each the set of data points whose initial influence estimate falls in one interval. For an optimal one-dimensional quantizer with M levels the share of the variance left inside the bins is about κ/M^2 [19, 9], with $\kappa = 2.7$ for normally distributed values, so a declared tolerance ϵ on that share gives $M_{\text{ref}} = \lceil \sqrt{2.7/\epsilon} \rceil$, 17 at $\epsilon = 0.01$. We start at $M_{\mathcal{I}} = M_{\text{ref}}$ and let the finite differences decide the rest. Each initial bin is differenced with the central stencil, two evaluations on N rows, giving U_k and $\Delta^2 T_k$; the U_k are centred so that $\sum_k p_k U_k = 0$.

Refinement splits the bins where the finite differences say the grouping is too coarse. Every bin with more than one point carries a proposed split and an expected gain, the increase in V_{btw} the split would produce if $\hat{\psi}_0$ were right, brought to the finite-differenced scale by ρ^2 , the ratio of V_{btw} to the between-bin variance of $\hat{\psi}_0$. The model's uncertainty inside a bin is v_k , the variance of its error around the error common to the bin (the common part is already fixed by U_k). The split is by level, a two-means on the bin's $\hat{\psi}_0$ values, when the estimate varies inside the bin more than v_k ; otherwise by adjacency, separating the points whose second-nearest prototype carries a higher prototype influence than their nearest, with expected gain $\rho^2 p_k v_k / N$. The bin with the largest expected gain is split; the smaller child is differenced with one forward step and the larger follows by subtraction, since a parent's bin influence is the mass-weighted sum of its children's. The realized gain is the increase in V_{btw} the split actually produced. A lineage stops after two consecutive splits whose realized gain falls below the bin's share of the tolerance, $\epsilon V_{\text{btw}} / L$ with L the current number of bins: one such split cannot tell a bin of constant influence from one whose variation was divided evenly between its children, a second can. The loop stops when no expected gain exceeds that share, and at most $1 + M_{\mathcal{X}}$ evaluations are spent refining. The ratio of realized to expected gains over the run is reported: it is how well $\hat{\psi}_0$ predicted. Refinement acts on variation that the initial estimate shows, or that

the model’s uncertainty inside a bin points to; Sect. 4.2 tests it on an influence that the initial estimate cannot see.

2.6 Outputs

Three quantities follow without a model. The between-bin term $V_{\text{btw}} = \frac{1}{N} \sum_k p_k U_k^2$ over the final L bins is finite-differenced on the full data, is a lower bound on V_{tot} exact up to the step error, and rises with every split. The bias term $\hat{B} = \frac{1}{2N} \sum_k p_k \Delta^2 T_k$ is the second-order term of the expansion of $\hat{T}$ along the bin masses; it is reported and not applied (Sect. 4.2 says why). The BCa acceleration $\hat{a}_{\text{BCa}}$ comes from the third moment of the refined estimate $\hat{\psi}(x_i) = U_{k(i)} + \rho(\hat{\psi}_0(x_i) - \bar{\psi}_{0,k(i)})$. The remainder is predicted: $\hat{V}_{\text{win}} = \rho^2 \frac{1}{N} \sum_k p_k \gamma_k [\text{Var}_k(\hat{\psi}_0) + v_k]$ over bins with more than one point, where $\gamma_k \leq 1$ is the realized fraction of the expected gain in the split that created the bin, so that a bin whose split showed the model’s variation to be spurious is discounted, and $\gamma_k = 1$ for a bin never split. $\hat{V}_{\text{tot}} = V_{\text{btw}} + \hat{V}_{\text{win}}$ is the variance estimate. The QIJ interval at level q , h_q^{QIJ} , is the BCa-form interval with variance $\hat{V}_{\text{tot}}$, acceleration $\hat{a}_{\text{BCa}}$, median-bias constant zero and centre $\hat{T}$. Reported alongside: evaluation counts by size, rows processed normalized by N , wall time, L and the split counts, ρ , the ratio of realized to expected gains, and the kernel width and noise ratio with a note when either sits on a bound.

2.7 Scope

Five conditions bound the method. The estimator must accept a weighted point set and be well defined on coarse weighted data within the data’s range. An evaluation that fails, or that ends on a bound of the estimator’s parameter box, returns no value; the method counts it, uses the same rule for a bootstrap replicate, and never retries. A prototype whose evaluation failed is left out of the influence model; a failed evaluation of an initial bin, or a prototype stage on which every prototype responds alike, fails the draw, and the paper reports how often that happened. An estimator that is not differentiable in the weights (a quantile, a hard threshold, model selection inside T) is out of scope, as for every influence-based method. An estimator whose influence is discontinuous in x , such as a tail probability, is in scope, with any loss reported: the prototype stage sees a jump only where a prototype sits on its far side, and a data point whose influence differs from its neighbours’ while its prototype’s centroid sits with the neighbours is invisible to the initial estimate. Whether the refinement recovers such a point depends on the model’s uncertainty marking its bin; where it does not, V_{btw} stays a lower bound short of V_{tot} by that point’s share and the interval under-covers by the same

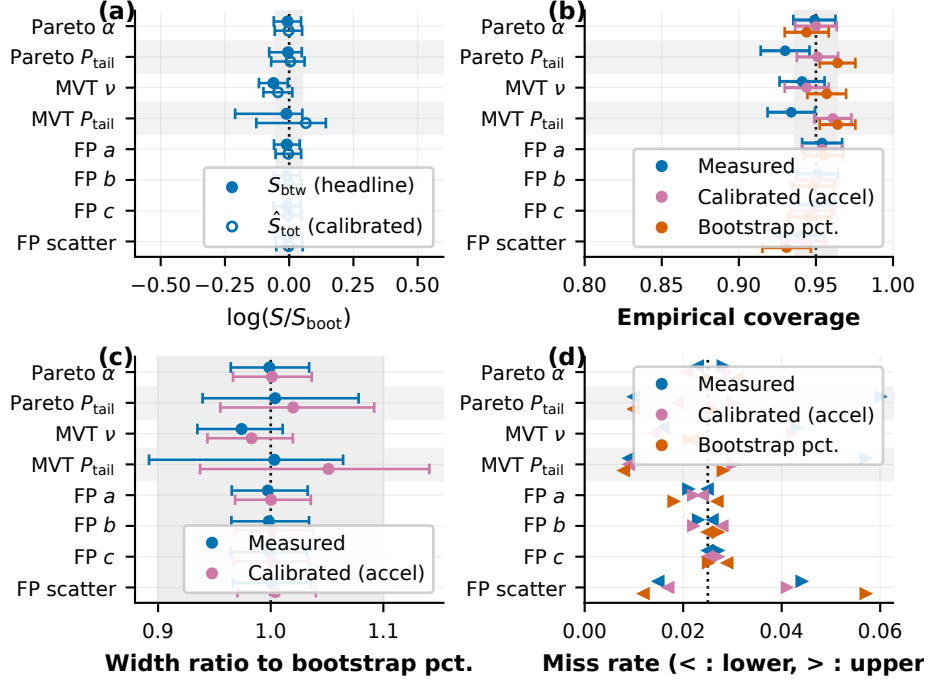


Fig. 1. The refined influence estimate $\hat{\psi}(x_i)$ against the true influence $\psi(x_i)$, one point per data point of one draw ($N = 2000$), with the diagonal as reference and the rank correlation printed. (a, b) multivariate t , v and S_{99} ; (c, d) Fundamental Plane, coefficient a and the scatter s ; (e, f) initial mass function, slope α and cutoff sharpness p . On S_{99} the points off the diagonal are tail points left inside mixed bins and their bin-mates. [Figure and the S_{99} correlation from the final build.]

share (Sect. 4.2). The value of T at the quantized data is the functional at that distribution and is never reported as an estimate of θ . The accuracy η is the user's declaration and is not checked. The multinomial covariance in (3) assumes iid sampling.

3 Experimental Data

We evaluate QIJ on four datasets. Two are from distributional families in which the ground truth is known; two are astronomical, with vector-valued estimands, and one of those is fitted by a constrained optimizer. Each estimator has an analytic influence function, which QIJ never sees: every estimator enters the method as a program that returns a value for a weighted dataset, and the closed form is used only to judge the result. From each data process, $S = 1000$ Monte Carlo draws of size

$N = 2000$ were taken. **QIJ** and the nonparametric bootstrap ($B = 2000$ replicates; “Boot” in the figures) were run on identical draws paired by seed. **QIJ**’s declared inputs were $\epsilon = 0.01$ throughout, and η at machine precision for the closed-form estimators, 10^{-12} for the two solved by a root-find or an eigendecomposition, and 10^{-6} for the initial mass function, the square root of its optimizer’s tolerance. The rule (5) then gave 188 prototypes for each one-output estimator, 371 for the Fundamental Plane and 414 for the initial mass function; every output started from 17 bins, and Table 1 reports where refinement ended. **Nothing was tuned per dataset.** The astronomical cases use an empirical population protocol: a large catalogue is treated as the population, draws are sampled from it with replacement, and the target θ_{true} is the parameter vector computed from the full catalogue. Below, we describe the datasets individually.

Pareto. A heavy-tailed distribution, with closed-form influence functions for both estimators (the shape parameter’s maximum-likelihood estimator, and the tail probability S_{99} , the survival function at the population’s 0.99 quantile, so that its true value is 0.01), so every stage of the pipeline can be checked analytically. **The shape parameter’s influence is unbounded in $\log x$, which makes the tail beyond the outermost prototype the test of the initial estimate; the tail probability’s influence is a step in one dimension, which the initial bins separate on their own.**

Multivariate t (MVT). A ten-dimensional, elliptically symmetric distribution with two estimands: the degrees of freedom ν (estimated by profile maximum likelihood), and the tail probability S_{99} beyond the population’s 0.99 quantile of radius. This tests whether our construction survives higher data dimension. **The ν estimator is the case in which the quantized data are far from the data: centroids thin the tails, and on the prototypes the estimator returns a ν in the hundreds for a $\nu = 5$ sample. The tail probability is the hard case: its influence is a step across a sphere in ten dimensions, and only about twenty of the 2000 points lie beyond it.**

Galaxy fundamental plane. The galaxy Fundamental Plane dataset (FP) comprises 76,997 early-type galaxies drawn from the Sloan Digital Sky Survey Data Release 7 spectroscopic catalog [1]. For each galaxy, the three Fundamental Plane observables are the aperture-corrected stellar velocity dispersion σ (following the correction of [15]), the effective radius R_e , and a luminosity-based surface brightness proxy I_e . The Fundamental Plane is fit in log-space as $\log R_e = a \log \sigma + b \log I_e + c$ via least squares. The estimands are the coefficient vector (a, b, c) and the intrinsic scatter s about the plane. **The four outputs come from one evaluation, so the Fundamental Plane is the vector case: one $\mathcal{X}$ -VQ and four $\mathcal{I}$ -VQs, the four sharing every evaluation.**

Stellar initial mass function (IMF). **The population is 19,857 stellar masses from the STARFORGE star-formation simulations [10, 11]. The esti-**

mator fits a two-regime mass function by weighted maximum likelihood: a gamma density below a fixed split of 0.177 solar masses, and above it a power law of slope α with an exponential cutoff at mass M_* whose sharpness is p . Continuity of the density and its slope at the split fixes the gamma’s shape and scale from (α, M_*, p) , so three parameters are free and five are reported; the same conditions tie α to the low-mass shape, so the numerous low-mass stars carry most of its sampling variance (Fig. 1(e)). The fit is a bounded trust-region minimization with an analytic gradient and Hessian, from one starting point computed by the method of moments from the weighted data, so that a bootstrap replicate and a QIJ evaluation are computed identically. A fit that ends on a bound of its parameter box returns no value and is counted as failed, for both methods. This is the case the method is meant for: one evaluation is an optimization, and the cutoff is identified by the few most massive stars, which a resample can lose.

4 Results

Table 1 reports every estimand; Figs. 1–3 show six of the thirteen, chosen to cover the four datasets and both kinds of difficulty. Every number in this section comes from the same 1000 draws of size 2000.

4.1 Accuracy Against the Truth and the Bootstrap

The reference for the variance is the linearized (oracle) variance from the analytic influence, and for coverage it is θ_{true} . On twelve of the thirteen estimands the between-bin term alone reaches 0.96–1.00 of the oracle variance, the predicted remainder adds under 0.6% of it, and the QIJ interval covers at 0.931–0.956 against the bootstrap’s 0.943–0.958 (standard error 0.007) with widths within 3% of the bootstrap’s on all but one (Fig. 2). Pooled over all thirteen estimands and all draws, the QIJ interval covers at 0.494, 0.899, 0.947 and 0.985 at nominal 0.50, 0.90, 0.95 and 0.99, within 0.3 percentage points of the bootstrap at every level. Fig. 1 shows what the second stage delivers: the refined estimate $\hat{\psi}$ against the true influence, point by point, with rank correlation 0.995 or above on the five smooth estimands shown, including the multivariate- t degrees of freedom, whose prototype influences were 250 times too large in scale ($\rho = 0.004$) and correctly ordered.

4.2 Where the Method Falls Short

[Tail paragraph under review: the author disputes that this is a limit of the method; a per-split trace is pending before it is settled.] The multivariate- t tail probability S_{99} is the one estimand where the interval under-covers: 0.934 against the bootstrap’s 0.953, with the between-bin term at a median 0.91 of the oracle variance (5–95% range 0.75–1.00

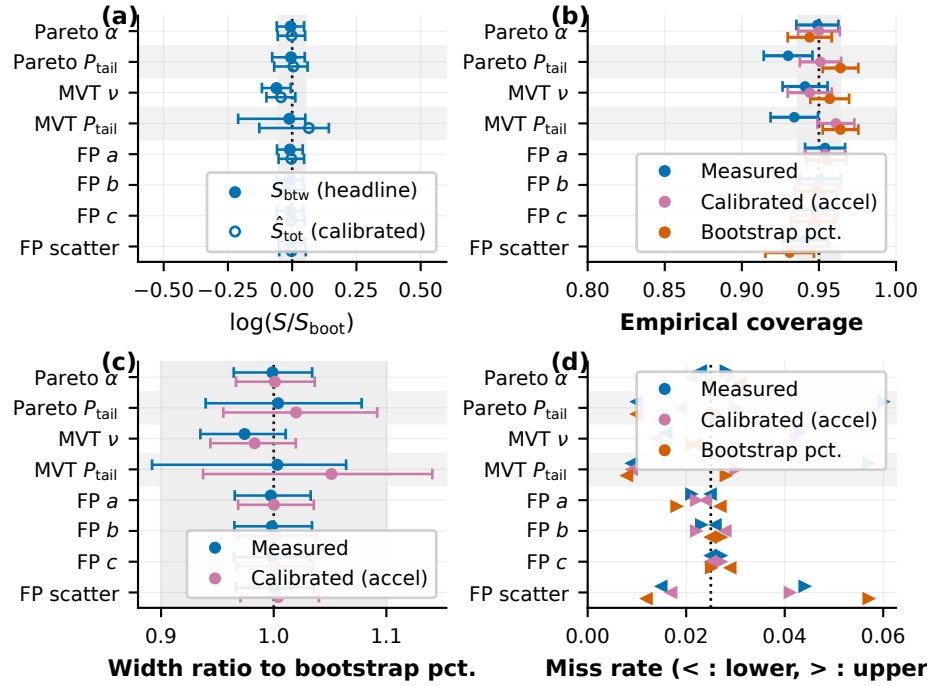


Fig. 2. QIJ against the bootstrap on the six featured estimands, rows in the order of Fig. 1. (a) Coverage at 0.95 of the QIJ and bootstrap intervals with Monte Carlo standard errors; the band is one standard error about the nominal line. (b) Width ratio, QIJ over bootstrap, median with the 5–95% range; the band marks $\pm 10\%$. Variance ratios for all thirteen estimands are in Table 1.

over draws) and a width 0.97 of the bootstrap's. The influence takes two values, about -0.01 inside a sphere and 0.99 beyond it, so the twenty or so points beyond the sphere carry the whole variance and each must end in a bin of its own. A point beyond the sphere whose receptive field holds two or three points inside it leaves the centroid inside, and that prototype's influence is the same as any interior one's: on the featured draw the prototype stage registered six of the seventeen such points, and the rest are invisible to the initial estimate, which is why the rank correlation in Fig. 1(b) is $[0.47]$. The refinement recovers what the model's uncertainty inside a bin points to, all of it on some draws and most of it on a typical one, at a median of 68 bins against 19–38 elsewhere. At $N = 1000$ the same estimand reached the oracle exactly, because k -means isolated the outlying points one per prototype; with denser data it isolates fewer. [Whether this is a limit of the method or of the stopping rule is open: on some draws the refinement recovers every such point, on others it stops with the

Table 1. All thirteen estimands over 1000 draws ($N = 2000$), medians over draws where a ratio. $V_{\text{btw}}/V_{\text{or}}$: the share of the oracle (linearized) variance measured directly; $\hat{V}_{\text{tot}}/V_{\text{or}}$ and $\hat{V}_{\text{tot}}/V_{\text{boot}}$: the variance estimate against the oracle and against the bootstrap; cov.: coverage at 0.95 of h^{QIJ} and h^{boot} (standard error 0.007); width: QIJ over bootstrap; L : final bins; evals: estimator evaluations per run; rows: normalized rows; time: QIJ over bootstrap wall time within a draw; failed: draws with no QIJ result / draws whose full-data fit failed (where the bootstrap fails too), of 1000.

	$\frac{V_{\text{btw}}}{V_{\text{or}}}$	$\frac{\hat{V}_{\text{tot}}}{V_{\text{or}}}$	$\frac{\hat{V}_{\text{tot}}}{V_{\text{boot}}}$	cov. QIJ	cov. boot	width	L	evals	rows	time	failed
Pareto α	0.996	1.000	0.81	0.951	0.948	1.00	23	230	59	0.80	0 / 0
Pareto S_{99}	1.000	1.000	0.89	0.944	0.951	1.01	19	226	55	0.90	0 / 0
MVT ν	0.993	0.997	0.97	0.951	0.950	0.99	24	231	60	0.36	3 / 0
MVT S_{99}	0.912	0.912	0.91	0.934	0.953	0.97	68	275	104	0.99	0 / 0
FP a	0.993	0.998	1.00	0.953	0.952	1.00	27	541	238	3.75	0 / 0
FP b	0.994	0.999	1.00	0.954	0.958	1.00	27	541	238	3.75	0 / 0
FP c	0.993	0.999	1.00	0.947	0.951	1.00	22	541	238	3.75	0 / 0
FP s	0.993	0.997	1.00	0.950	0.948	1.00	23	541	238	3.75	0 / 0
IMF α	0.963	0.965	0.95	0.947	0.955	0.97	38	663	334	0.31	31 / 17
IMF M_*	0.996	0.999	1.00	0.931	0.944	1.01	25	663	334	0.31	31 / 17
IMF p	0.964	0.967	0.10	0.956	0.951	0.66	33	663	334	0.31	31 / 17
IMF γ shape	0.995	0.996	0.99	0.947	0.943	1.00	32	663	334	0.31	31 / 17
IMF γ scale	0.994	0.996	0.99	0.944	0.944	1.00	33	663	334	0.31	31 / 17

between-bin term short by the remaining share.] The method’s own diagnostic flags it: the ratio of realized to expected gains is above 1.5 on 95% of draws (median 3.2), meaning the splits kept finding more variance than the model predicted, whereas on every other estimand the median lies between 0.4 and 1.6 or no split is needed at all.

The initial mass function’s cutoff mass M_* is the other shortfall: the QIJ interval covers at 0.931 against the bootstrap’s 0.944. QIJ’s variance for M_* equals the oracle’s, which is a linearization at the fitted parameters; the Monte Carlo sampling distribution of $\hat{M}_*$ at $N = 2000$ is 12% wider in standard deviation and left-skewed. Both methods see a finite-sample effect the linearization does not[, and Fig. 4 shows it vanishing with N]. The bias term does not help: over the draws the realized bias of $\hat{M}_*$ is zero within error while the bias term is heavy-tailed from draw to draw, and shifting the interval by it in either direction lowers coverage. This is why the bias term is reported and not applied. The other IMF outputs show the case the method is meant for. The cutoff sharpness p is identified by the few most massive stars, and 7.8% of bootstrap replicates lose enough of them that the fit runs to a bound and returns nothing, while 1.6% of QIJ’s evaluations do; the bootstrap’s interval for p is 1.5 times the width of QIJ’s at the same coverage (0.951 against 0.956). Of the 1000 draws, 31 failed for QIJ (17 at the full-data fit, where the bootstrap fails too, and 14 in the prototype stage) and are counted, not dropped.

4.3 Cost

Cost is reported in three forms: evaluation counts by the number of rows each processes, rows processed normalized by N (one full-data evaluation equals one normalized row), and wall time. The QIJ interval costs 55–60 normalized rows for a one-output closed-form estimator, 104 for the multivariate- t tail, 238 for the four-output Fundamental Plane and 334 for the five-output initial mass function, against 2000 full-data evaluations for the bootstrap. Wall time follows the evaluation count only where an evaluation is expensive: on the initial mass function, where one evaluation is an optimization, a QIJ run takes 64 s against the bootstrap’s 214 s, and on the multivariate- t degrees of freedom 0.36 of the bootstrap’s time; on the closed-form estimators the two are comparable, and on the Fundamental Plane, whose evaluation is one eigendecomposition, QIJ’s own overhead (the Gaussian process and its per-point uncertainty) makes it 3.7 times slower than the bootstrap. Fig. 3 shows, per featured estimand, the bootstrap’s relative width error after its first b replicates against b , and the QIJ interval at the normalized rows it spent. Fig. 4 shows the same on the Fundamental Plane as N grows to the full catalogue[: numbers from the cost sweep].

5 Conclusions and Future Work

We have shown that a fitted vector quantizer supports nonparametric interval estimation for an estimator whose influence function has no closed form. The route is not resampling from the partition, which would require the sampling variability that the quantizer minimizes away, but differentiation across it. The quantizer’s prototypes give a cheap estimate of the influence; a second quantizer of that estimate gives bins inside which the influence varies little; finite differences along the bin masses measure the between-bin variance on the full data, and the same differences say where to refine. The decomposition of (4) is an algebraic identity on any dataset and any partition, and it is what lets the method report how much of the variance it measured.

On twelve of thirteen estimands across four datasets the QIJ interval matches the bootstrap’s coverage, with 55 to 334 full-data-evaluation equivalents against the bootstrap’s 2000; where one evaluation is an optimization that is a third of the bootstrap’s wall time, and the method also avoids the bound hits that resampling causes there. The two shortfalls are explained by the construction. A tail probability in ten dimensions puts all of its variance on a handful of points, and a point that is not its own prototype is invisible to the initial estimate; the refinement recovers most of it, the between-bin term remains a lower bound, and the interval under-covers by the remainder. A cutoff mass identified by rare stars has a finite-sample spread

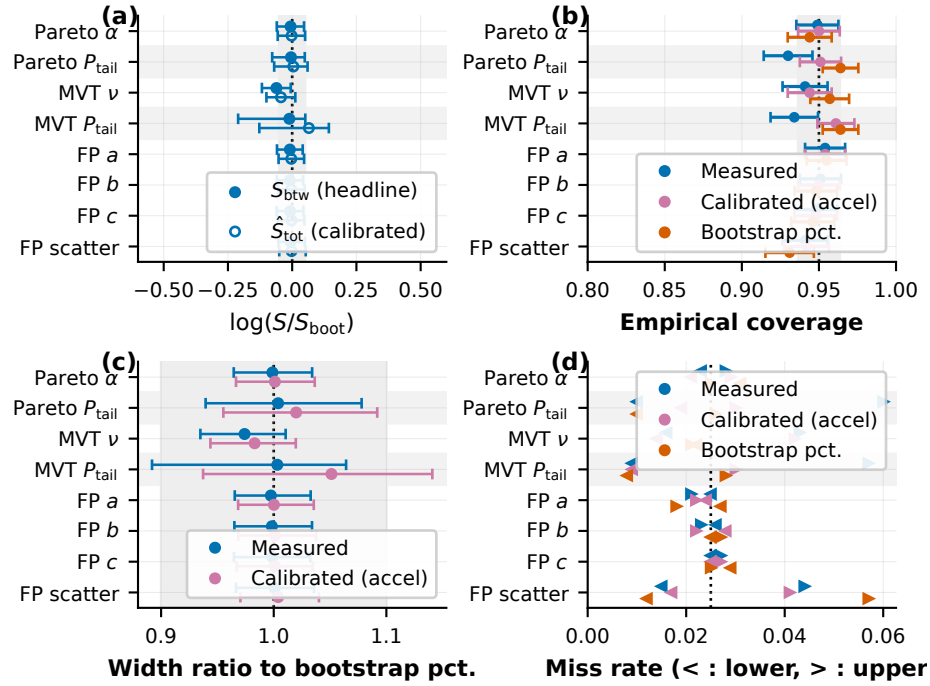


Fig.3. Cost on the six featured estimands. Curve: the bootstrap’s relative width error at 0.95 after the first b replicates, mean over draws with the interquartile band. Marker: QIJ at its median normalized rows with interquartile whiskers, its width error against the converged bootstrap. Printed: wall time of one run of each method, one worker, one thread. Both Fundamental Plane panels print the same pair, since its four outputs come from one run.

that a linearized variance does not see, and neither the bootstrap nor QIJ covers it at nominal at this sample size. Both are reported by the method’s own diagnostics.

The prototype count is set by a cost budget, not by accuracy, and no rule known before running can promise that the prototypes resolve an unseen estimator’s influence. Two directions follow. A refinement that spends evaluations to search for variance the model does not predict, rather than to measure splits the model proposes, would address the tail case. And a quantizer prescribed for inference rather than compression, which places prototypes where the influence changes rather than where the data are dense, would move the first stage’s blind spot.

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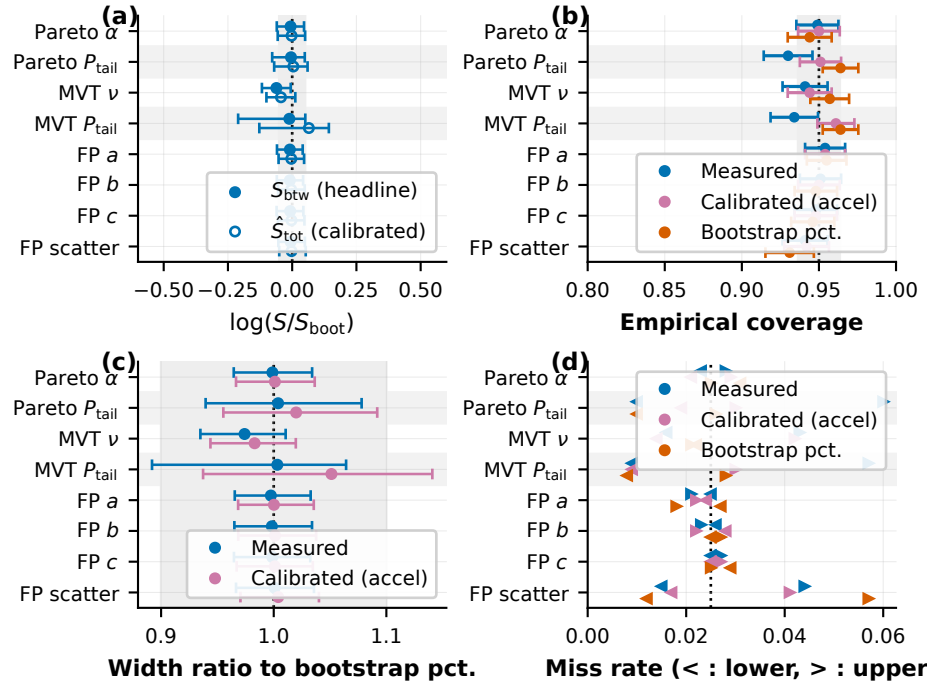


Fig. 4. Cost against sample size on the Fundamental Plane, N from 1000 to the full catalogue of 76,997 galaxies. (a) Normalized rows and wall time per run for QIJ and the bootstrap. (b) Coverage of both intervals at 0.95, one line per output with Monte Carlo bands, and the width ratio. [IMF panels if shown.]

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References

1. Abazajian, K.N., et al.: The seventh data release of the Sloan Digital Sky Survey. The Astrophysical Journal Supplement Series **182**(2), 543–558 (2009)
2. Bickel, P.J., Götze, F., van Zwet, W.R.: Resampling fewer than n observations: gains, losses, and remedies for losses. Statistica Sinica **7**(1), 1–31 (1997)
3. DiCiccio, T.J., Efron, B.: Bootstrap confidence intervals. Statistical Science **11**(3), 189–228 (1996)
4. Efron, B.: Bootstrap methods: another look at the jackknife. The Annals of Statistics **7**(1), 1–26 (1979)

5. Efron, B.: The Jackknife, the Bootstrap and Other Resampling Plans. No. 38 in CBMS-NSF Regional Conference Series in Applied Mathematics, SIAM, Philadelphia (1982)
6. Efron, B.: Better bootstrap confidence intervals. *Journal of the American Statistical Association* **82**(397), 171–185 (1987)
7. Feldman, D., Langberg, M.: A unified framework for approximating and clustering data. In: *Proceedings of the 43rd ACM Symposium on Theory of Computing (STOC)*. pp. 569–578 (2011)
8. Giordano, R., Stephenson, W., Liu, R., Jordan, M.I., Broderick, T.: A Swiss army infinitesimal jackknife. In: *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*. PMLR, vol. 89, pp. 1139–1147 (2019)
9. Graf, S., Luschgy, H.: *Foundations of Quantization for Probability Distributions*. No. 1730 in *Lecture Notes in Mathematics*, Springer, Berlin (2000)
10. Grudić, M.Y., Guszejnov, D., Hopkins, P.F., Offner, S.S.R., Faucher-Giguère, C.A.: STARFORGE: Towards a comprehensive numerical model of star cluster formation and feedback. *Monthly Notices of the Royal Astronomical Society* **506**(2), 2199–2231 (2021)
11. Guszejnov, D., Grudić, M.Y., Offner, S.S.R., Faucher-Giguère, C.A., Hopkins, P.F., Rosen, A.L.: Effects of the environment and feedback physics on the initial mass function of stars in the STARFORGE simulations. *Monthly Notices of the Royal Astronomical Society* **515**(4), 4929–4952 (2022)
12. Hammer, B., Villmann, T.: Generalized relevance learning vector quantization. *Neural Networks* **15**(8–9), 1059–1068 (2002)
13. Hampel, F.R.: The influence curve and its role in robust estimation. *Journal of the American Statistical Association* **69**(346), 383–393 (1974)
14. Jaeckel, L.A.: The infinitesimal jackknife. Memorandum MM 72-1215-11, Bell Laboratories, Murray Hill, NJ (1972)
15. Jørgensen, I., Franx, M., Kjærgaard, P.: Spectroscopy for E and S0 galaxies in nine clusters. *Monthly Notices of the Royal Astronomical Society* **276**(4), 1341–1364 (1995)
16. Kleiner, A., Talwalkar, A., Sarkar, P., Jordan, M.I.: A scalable bootstrap for massive data. *Journal of the Royal Statistical Society: Series B* **76**(4), 795–816 (2014)
17. Koh, P.W., Liang, P.: Understanding black-box predictions via influence functions. In: *Proceedings of the 34th International Conference on Machine Learning (ICML)*. PMLR, vol. 70, pp. 1885–1894 (2017)
18. von Mises, R.: On the asymptotic distribution of differentiable statistical functions. *The Annals of Mathematical Statistics* **18**(3), 309–348 (1947)
19. Panter, P.F., Dite, W.: Quantization distortion in pulse-count modulation with nonuniform spacing of levels. *Proceedings of the IRE* **39**(1), 44–48 (1951)
20. Rasmussen, C.E., Williams, C.K.I.: *Gaussian Processes for Machine Learning*. MIT Press, Cambridge, MA (2006)
21. Schneider, P., Biehl, M., Hammer, B.: Adaptive relevance matrices in learning vector quantization. *Neural Computation* **21**(12), 3532–3561 (2009)
22. Stefanski, L.A., Boos, D.D.: The calculus of M-estimation. *The American Statistician* **56**(1), 29–38 (2002)
23. Wager, S., Hastie, T., Efron, B.: Confidence intervals for random forests: the jackknife and the infinitesimal jackknife. *Journal of Machine Learning Research* **15**, 1625–1651 (2014)

24. Xia, Y., Tong, H., Li, W.K., Zhu, L.X.: An adaptive estimation of dimension reduction space. *Journal of the Royal Statistical Society: Series B* **64**(3), 363–410 (2002)